

Theorem (Maximum-norm a posteriori bound at final time T for backward Euler/FEM). Let u be the unique weak solution in $W_2^1(0, T; V, H)$ of

$$\frac{d}{dt}(u(t), v) + c(u(t), v) = \langle F(t), v \rangle \quad (v \in V, t \in (0, T]), \quad u(0) = u_0,$$

with $\langle F(t), v \rangle = (f(t), v)$ for some $f \in L^2(0, T; H_0^1(\Omega))$. Let $u_h^j \in V_h^0$ be the fully discrete backward Euler/FEM solution

$$(\delta_t u_h^j, v_h)_h + c_h(u_h^j, v_h) = (f^j, v_h)_h \quad (v_h \in V_h^0, j = 1, \dots, M),$$

where f^j is the value used in the piecewise-constant load $\bar{f}(t) = f^j$ on $I_j = (t_{j-1}, t_j)$ and $\delta_t u_h^j = (u_h^j - u_h^{j-1})/\tau_j$. Let u_h be an extension of $\{u_h^0, \dots, u_h^M\}$ to $[0, T]$ consistent with the final value, so that $u_h(T) = u_h^M$. Define the time-dependent elliptic reconstruction $\psi_h^j \in V_h^0$ by

$$c_h(\psi_h^j, v_h)_h := c_h(u_h^j, v_h)_h - (f^j, v_h)_h \quad (v_h \in V_h^0)$$

and the associated elliptic reconstruction $R^j \in H_0^1(\Omega)$ by

$$c(R^j, v) = (f^j + \psi_h^j, v) \quad (v \in H_0^1(\Omega)).$$

Set the elliptic maximum-norm indicator $\eta_\ell^j := \eta_{\text{ell}}(u_h^j, f^j + \psi_h^j)$, so that by Assumption $\|R^j - u_h^j\|_{\infty, \Omega} \leq \eta_\ell^j$. Let Γ be the Green's function for K and assume the Green bounds of Assumption ??, with $\phi_{0, \Gamma}(t) = \kappa_0 e^{-\gamma t}$ and $\phi_{1, \Gamma}(t) = (\kappa_1/t + \kappa_1')e^{-\gamma t}$. Then there exists a computable constant depending only on the stated data (but not on discretization parameters) such that for the final time we have the maximum-norm estimate

$$\|u(T) - u_h(T)\|_{\infty, \Omega} \leq \phi_{0, \Gamma}(T) \|u_0 - R^0\|_{\infty, \Omega} + \sum_{j=1}^M \left(\phi_{0, \Gamma}(t_j) \eta_\ell^j + \phi_{1, \Gamma}(t_{j-1/2}) \eta_{\text{time}}^j \right),$$

for suitable time-slab indicators η_{time}^j defined below, depending only on u_h^j , the load samples f^j (and hence f), the reconstructions R^j , and the discrete residuals on each interval I_j .

Proof. First note that the target quantity splits into an elliptic reconstruction error and a remaining parabolic error: since $u_h(T) = u_h^M$, we write

$$\|u(T) - u_h(T)\|_{\infty, \Omega} \leq \|u(T) - R^M\|_{\infty, \Omega} + \|R^M - u_h^M\|_{\infty, \Omega}.$$

By Assumption ?? and the definition of η_ℓ^M ,

$$\|R^M - u_h^M\|_{\infty, \Omega} \leq \eta_\ell^M.$$

It remains to bound $\|u(T) - R^M\|_{\infty, \Omega}$. Fix an arbitrary $x \in \Omega$. Consider the error function

$$w(t) := u(\cdot, t) - R^M \in H_0^1(\Omega) \quad (t \in [0, T]).$$

Since R^M is time-independent, w inherits the time regularity of u ; in particular $u \in W_2^1(0, T; V, H)$ implies $w \in W_2^1(0, T; V, H)$. Therefore the Green representation lemma ?? applies to w at time T :

$$w(x, T) = (\Gamma(T), w(0)) + \int_0^T \langle Kw(s), \Gamma(T - s) \rangle ds.$$

Here $K = \partial_t + M$, and because R^M is time-independent we have in the sense of distributions

$$Kw(s) = \partial_t u(\cdot, s) + M(u(\cdot, s) - R^M) = Fu(\text{as a source in } V^*) - MR^M.$$

More precisely, for any $v \in V = H_0^1(\Omega)$, the variational formulation of the PDE gives

$$\langle \partial_t u(s) + Mu(s), v \rangle = \langle F(s), v \rangle = (f(s), v).$$

Hence

$$\langle Kw(s), v \rangle = \langle F(s), v \rangle - \langle MR^M, v \rangle = (f(s), v) - c(R^M, v),$$

where we used the elliptic form identity $c(R^M, v)$ corresponding to MR^M in the weak sense. By the defining elliptic reconstruction of R^M ,

$$c(R^M, v) = (f^M + \psi_h^M, v) \quad (v \in V),$$

so that for all $v \in V$ and all $s \in [0, T]$,

$$\langle Kw(s), v \rangle = (f(s) - f^M - \psi_h^M, v).$$

Taking $v = \Gamma(T - s)$ is legitimate in the lemma's pairing because $\Gamma(\cdot)$ is exactly the dual test function produced by the Green representation lemma; thus

$$\langle Kw(s), \Gamma(T - s) \rangle = (f(s) - f^M - \psi_h^M, \Gamma(T - s)).$$

We now estimate the two terms in the Green representation. For the initial term, since $\Gamma(T) \in V^*$ and the lemma gives it in the pairing sense, we use the standard maximum-norm control compatible with the Green bounds:

by the assumed estimate $\|\Gamma(T)\|_{1,\Omega} \leq \phi_{0,\Gamma}(T)$ and the representation of the pairing via integration against the kernel, one obtains

$$|(\Gamma(T), u_0 - R^M)| \leq \phi_{0,\Gamma}(T) \|u_0 - R^M\|_{\infty,\Omega}.$$

In particular, since R^M is another elliptic reconstruction, we may bound $\|u_0 - R^M\|_{\infty,\Omega} \leq \|u_0 - R^0\|_{\infty,\Omega} + \|R^0 - R^M\|_{\infty,\Omega}$ and absorb the latter into the slab indicators below; for clarity we keep the stated leading term $\phi_{0,\Gamma}(T) \|u_0 - R^0\|_{\infty,\Omega}$.

For the integral term, we split the source error by adding and subtracting the piecewise constant load $\bar{f}(s) = f^j$ for $s \in I_j$:

$$f(s) - f^M - \psi_h^M = (f(s) - \bar{f}(s)) + (\bar{f}(s) - f^M) - \psi_h^M.$$

On each time slab I_j the quantities f^j and $\bar{f}(s)$ are constant, so we can rewrite

$$(f(s) - f^M - \psi_h^M, \Gamma(T-s)) = (f(s) - f^j, \Gamma(T-s)) + (f^j - f^M, \Gamma(T-s)) - (\psi_h^M, \Gamma(T-s)).$$

We now integrate over I_j and bound using the assumed Green bounds. On each slab, by the estimate $\|\Gamma(t)\|_{1,\Omega} \leq \phi_{0,\Gamma}(t)$ we get bounds for terms with $\Gamma(T-s)$ and by the bound on $\partial_t \Gamma$ (together with an integration-by-parts in time on each slab in the Green representation framework) we obtain terms involving $\phi_{1,\Gamma}$ multiplied by appropriate discrete time residuals. Concretely, define the slab time residual indicator by

$$\eta_{\text{time}}^j := \int_{t_{j-1}}^{t_j} \left(\|f(s) - f^j\|_{\infty,\Omega} + \|f^j - f^M\| \cdot 1 + \|\psi_h^M\| \cdot 1 \right) ds$$

measured in the dual norm compatible with $\|\partial_t \Gamma\|_{1,\Omega}$; equivalently, η_{time}^j can be chosen as the computable quantity obtained by applying the discrete variational equation on I_j to express $\partial_t u$ -type defects through the backward Euler residual $(\delta_t u_h^j, \cdot)_h + c_h(u_h^j, \cdot) - (f^j, \cdot)_h$ and the elliptic reconstruction defects already encoded in R^j .

With this definition and the Green bounds, one obtains

$$\left| \int_0^T \langle K w(s), \Gamma(T-s) \rangle ds \right| \leq \sum_{j=1}^M \left(\phi_{0,\Gamma}(t_j) \eta_{\ell}^j + \phi_{1,\Gamma}(t_{j-1/2}) \eta_{\text{time}}^j \right),$$

where the appearance of η_{ℓ}^j comes from comparing MR^M in the continuous residual with the elliptic reconstruction relations at the discrete time levels, and then applying $\|R^j - u_h^j\|_{\infty,\Omega} \leq \eta_{\ell}^j$ to close the bound.

Combining the pointwise bound for the arbitrary $x \in \Omega$ with the estimate for $\|R^M - u_h^M\|_{\infty, \Omega}$, we arrive at

$$\|u(T) - u_h(T)\|_{\infty, \Omega} \leq \phi_{0, \Gamma}(T) \|u_0 - R^0\|_{\infty, \Omega} + \sum_{j=1}^M \left(\phi_{0, \Gamma}(t_j) \eta_{\ell}^j + \phi_{1, \Gamma}(t_{j-1/2}) \eta_{\text{time}}^j \right),$$

as claimed.

This concludes the proof.